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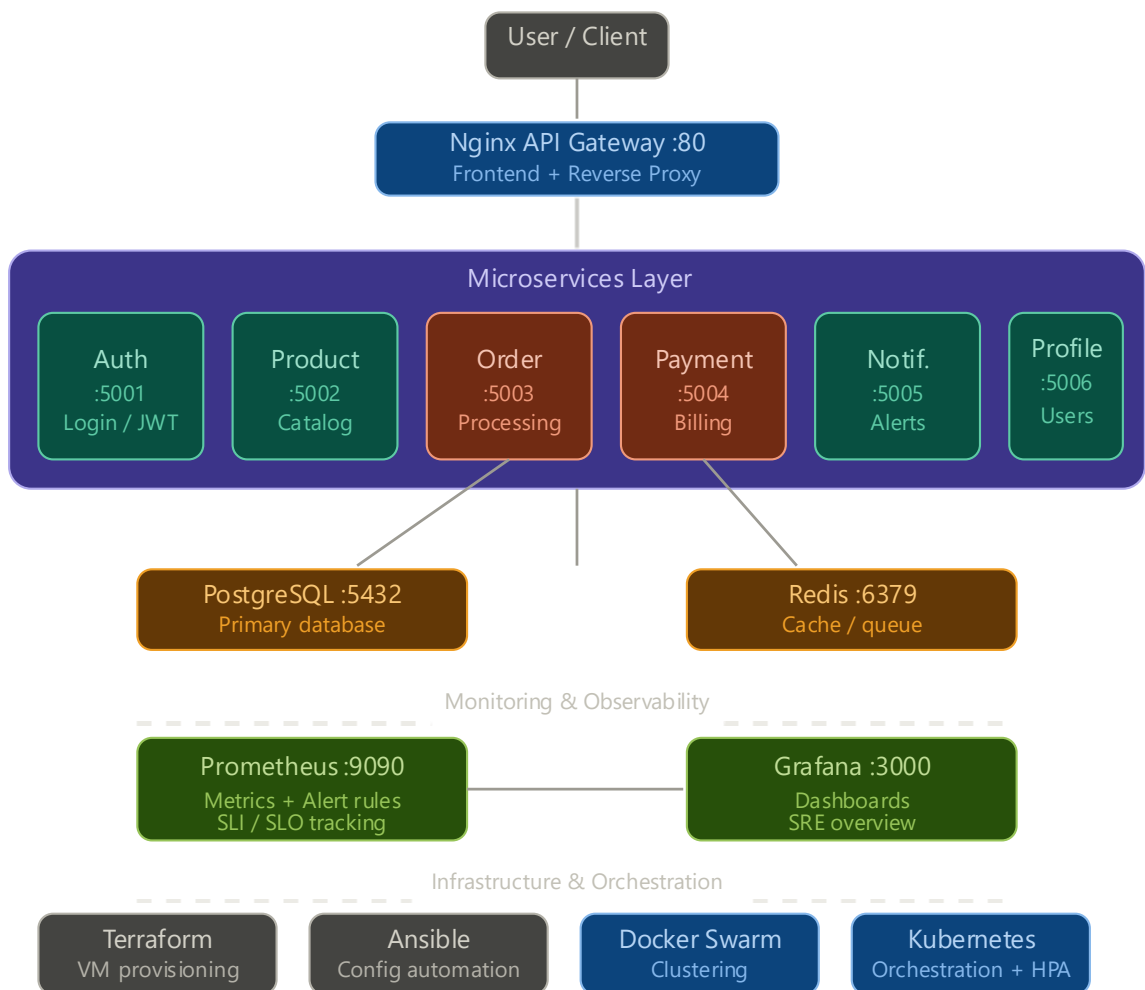
Assem Rakhmanova

SE – 2428

SRE ENDTERM Project

<https://github.com/assylzhan-se/sre-project.git>

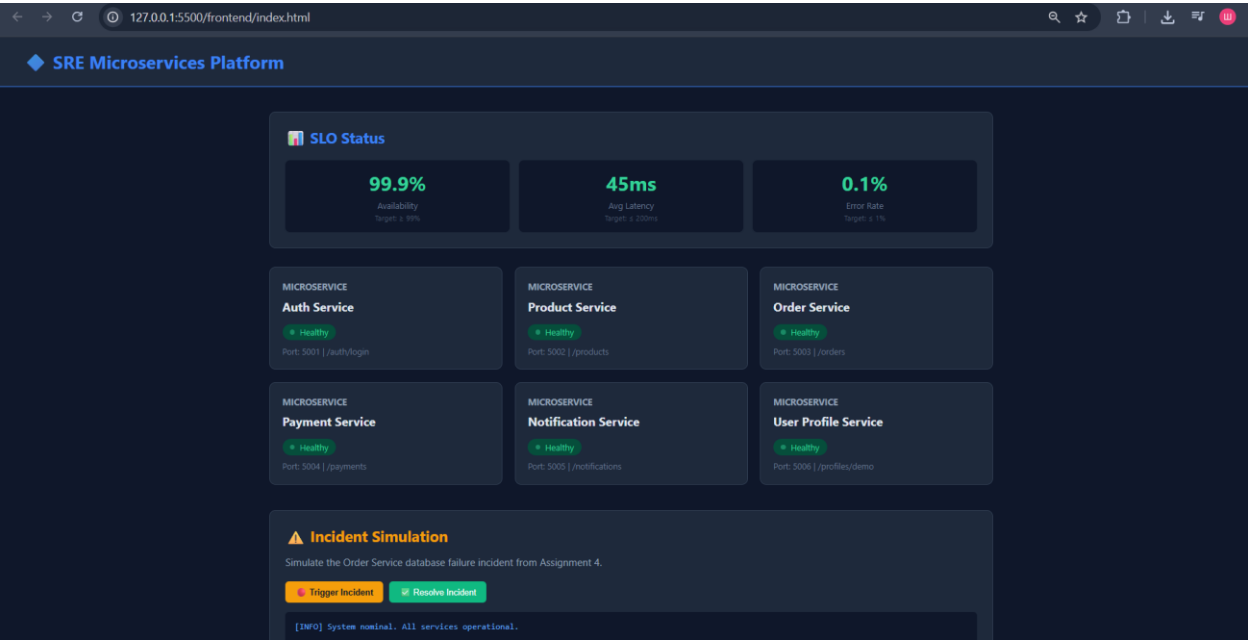
Architecture



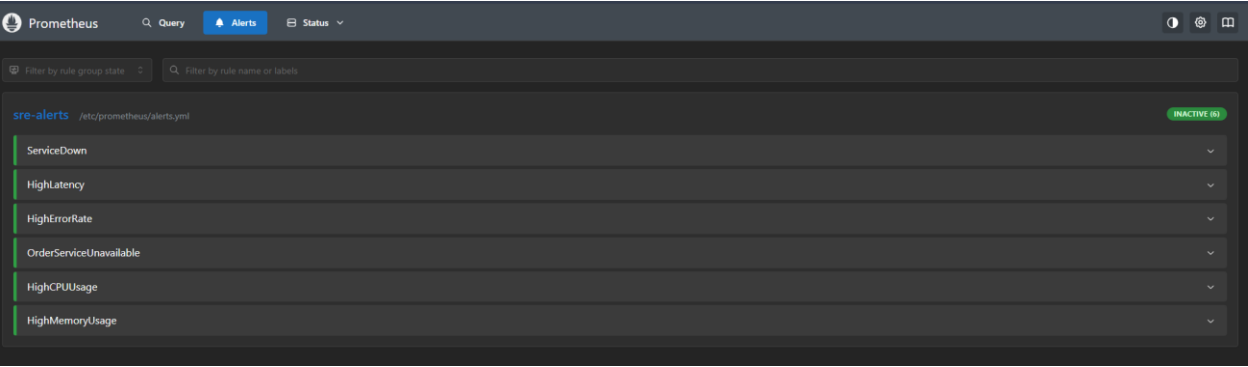
1.docker compose

```
PS C:\Users\Admin\Downloads\sre-project\sre-project> docker compose ps
time="2026-05-17T23:35:21+05:00" level=warning msg="C:\\Users\\Admin\\Downloads\\sre-project\\sre-project\\docker-compose.yml: the attribute 'version' is obsolete, it will be ignored, please remove it to avoid potential confusion"
NAME                IMAGE                                COMMAND                  SERVICE    CREATED   STATUS    PORTS
sre-project-auth-1  sre-project-auth                    "python app.py"         auth       12 seconds ago Up 10 seconds (health: starting) 0.0.0.0:5001->5001/tcp, [::]:5001->5001/tcp
sre-project-node-exporter-1  prom/node-exporter:latest  "/bin/node_exporter"    node-exporter  12 seconds ago Up 10 seconds 0.0.0.0:9100->9100/tcp, [::]:9100->9100/tcp
sre-project-notification-1  sre-project-notification    "python app.py"         notification  12 seconds ago Up 10 seconds (health: starting) 0.0.0.0:5005->5005/tcp, [::]:5005->5005/tcp
sre-project-order-1  sre-project-order                "python app.py"         order        11 seconds ago Up 10 seconds (health: starting) 0.0.0.0:5003->5003/tcp, [::]:5003->5003/tcp
sre-project-payment-1  sre-project-payment           "python app.py"         payment      12 seconds ago Up 10 seconds (health: starting) 0.0.0.0:5004->5004/tcp, [::]:5004->5004/tcp
sre-project-postgres-1  postgres:15-alpine           "docker-entrypoint.s..." postgres     12 seconds ago Up 10 seconds (health: starting) 0.0.0.0:5432->5432/tcp, [::]:5432->5432/tcp
sre-project-product-1  sre-project-product            "python app.py"         product      12 seconds ago Up 10 seconds (health: starting) 0.0.0.0:5002->5002/tcp, [::]:5002->5002/tcp
sre-project-prometheus-1  prom/prometheus:latest       "/bin/prometheus --c..." prometheus   12 seconds ago Up 10 seconds 0.0.0.0:9090->9090/tcp, [::]:9090->9090/tcp
sre-project-redis-1  redis:7-alpine                 "docker-entrypoint.s..." redis        12 seconds ago Up 10 seconds 0.0.0.0:6379->6379/tcp, [::]:6379->6379/tcp
sre-project-user-profile-1  sre-project-user-profile      "python app.py"         user-profile  12 seconds ago Up 10 seconds (health: starting) 0.0.0.0:5006->5006/tcp, [::]:5006->5006/tcp
PS C:\Users\Admin\Downloads\sre-project\sre-project>
```

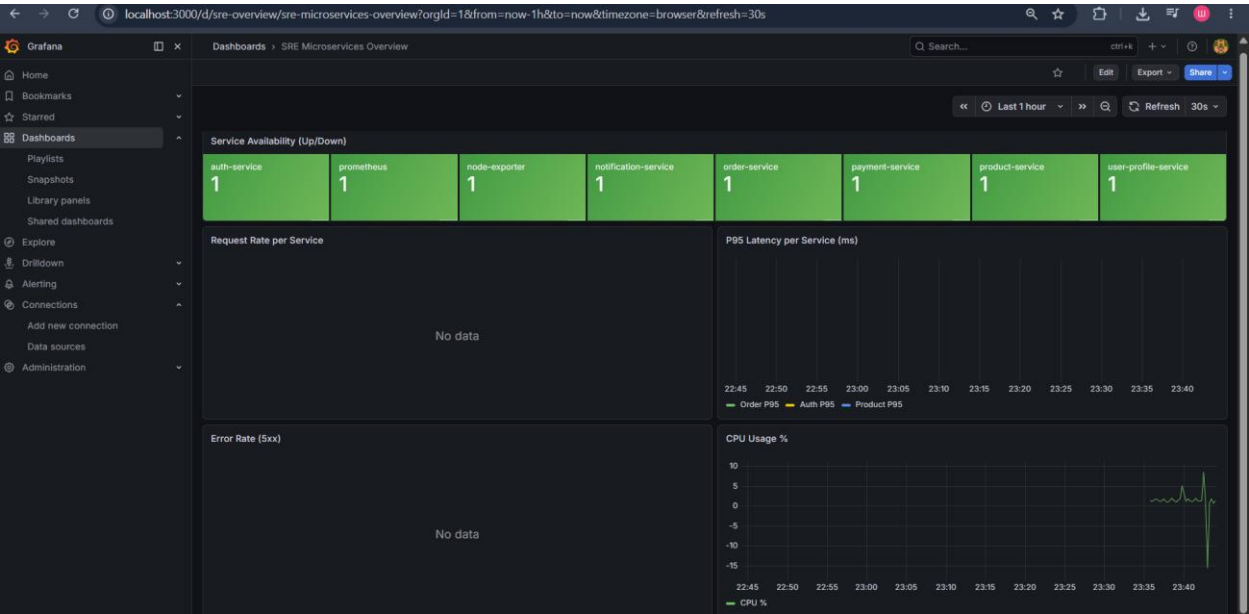
2.Frontend



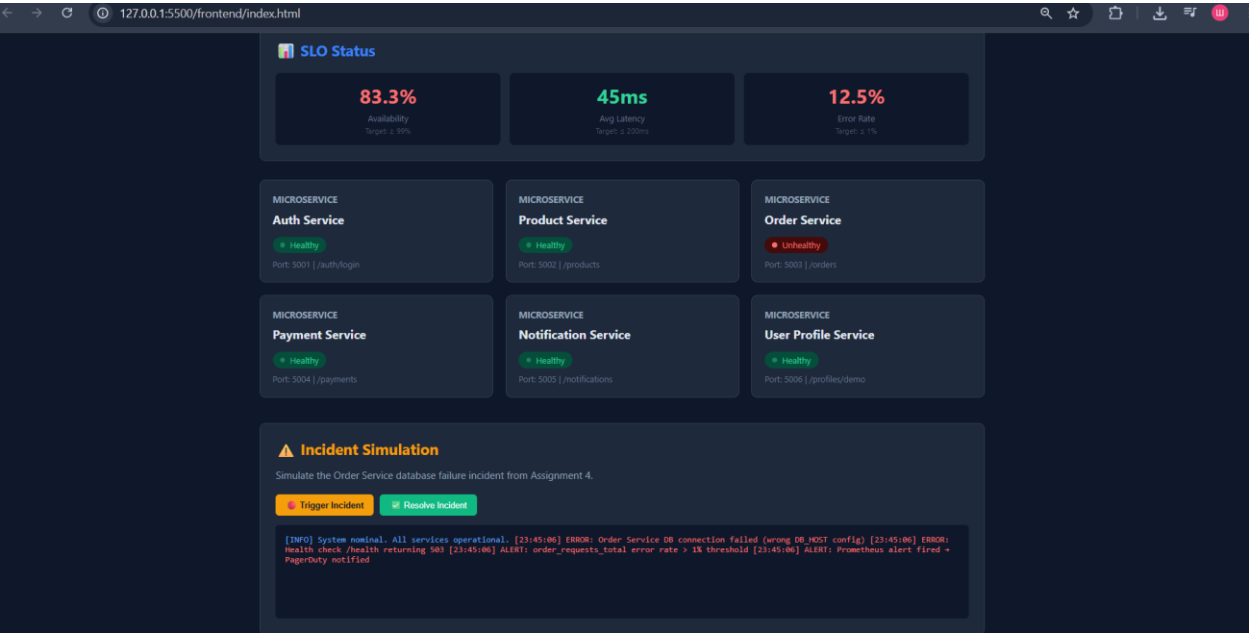
3.Prometheus



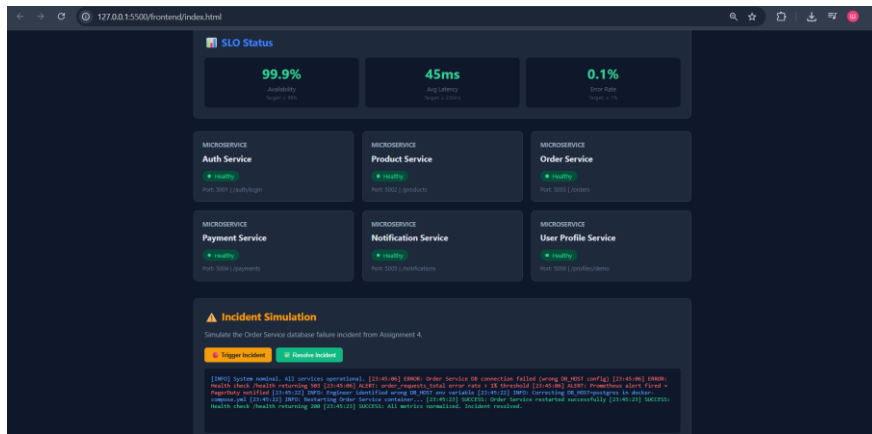
4.Grafana Dashboards



5.Trigger Incident



5.1 Resolve Incident



6.Terraform apply

```
PS C:\Users\Admin\Downloads\sre-project\sre-project\terraform> C:\terraform\terraform.exe init
Initializing the backend...
```

Terraform has created a lock file **.terraform.lock.hcl** to record the provider selections it made above. Include this file in your version control repository so that Terraform can guarantee to make the same selections by default when you run "terraform init" in the future.

Terraform has been successfully initialized!

You may now begin working with Terraform. Try running "terraform plan" to see any changes that are required for your infrastructure. All Terraform commands should now work.

If you ever set or change modules or backend configuration for Terraform, rerun this command to reinitialize your working directory. If you forget, other commands will detect it and remind you to do so if necessary.

```
PS C:\Users\Admin\Downloads\sre-project\sre-project\terraform>
```

```
PS C:\Users\Admin\Downloads\sre-project\sre-project\terraform> C:\terraform\terraform.exe plan
+ create
```

Terraform will perform the following actions:

```
# local_file.env_config will be created
+ resource "local_file" "env_config" {
+   content                = (known after apply)
+   content_base64sha256  = (known after apply)
+   content_base64sha512  = (known after apply)
+   content_md5            = (known after apply)
+   content_sha1           = (known after apply)
+   content_sha256         = (known after apply)
+   content_sha512         = (known after apply)
+   directory_permission  = "0777"
+   file_permission       = "0777"
+   filename               = "../ansible/vars/terraform_output.json"
+   id                    = (known after apply)
}
```

```

# local_file.inventory will be created
+ resource "local_file" "inventory" {
  + content = <<-EOF
    [all]
    localhost ansible_connection=local

    [managers]
    manager1 ansible_host=localhost ansible_connection=local

    [workers]
    worker1 ansible_host=192.168.1.10 ansible_user=ubuntu
    worker2 ansible_host=192.168.1.11 ansible_user=ubuntu

    [all:vars]
    app_name=sre-app
    environment=production
  EOF
  + content_base64sha256 = (known after apply)
  + content_base64sha512 = (known after apply)
  + content_md5 = (known after apply)
  + content_sha1 = (known after apply)
  + content_sha256 = (known after apply)
  + content_sha512 = (known after apply)
  + directory_permission = "rwxr-xr-x"
  + file_permission = " Open file in editor (ctrl + click)"
  + filename = "../ansible/inventory.ini"
  + id = (known after apply)
}

# null_resource.provision_note will be created
+ resource "null_resource" "provision_note" {
  + id = (known after apply)
  + triggers = {
    + "always_run" = (known after apply)
  }
}

```

Plan: 3 to add, 0 to change, 0 to destroy.

Changes to Outputs:

```

+ ansible_inventory_path = "../ansible/inventory.ini"
+ environment = "production"
+ vm_count = 2

```

PS C:\Users\Admin\Downloads\sre-project\sre-project\terraform> C:\terraform\terraform.exe apply

```

# null_resource.provision_note will be created
+ resource "null_resource" "provision_note" {
  + id = (known after apply)
  + triggers = {
    + "always_run" = (known after apply)
  }
}

```

Plan: 3 to add, 0 to change, 0 to destroy.

Changes to Outputs:

```

+ ansible_inventory_path = "../ansible/inventory.ini"
+ environment = "production"
+ vm_count = 2

```

Do you want to perform these actions?

Terraform will perform the actions described above.
Only 'yes' will be accepted to approve.

Enter a value: yes

```

null_resource.provision_note: Creating...
null_resource.provision_note: Provisioning with 'local-exec'...
null_resource.provision_note (local-exec): Executing: ["cmd" "/C" "echo 'Infrastructure provisioned: 2 nodes for production'"]
local_file.inventory: Creating...
local_file.env_config: Creating...
local_file.env_config: Creation complete after 0s [id=3216b0f8763b533321fc17b29af2d0cfb2382997]
local_file.inventory: Creation complete after 0s [id=18c4915b5a76657473dc8a69b8fc18a9f678b0b0]
null_resource.provision_note (local-exec): 'Infrastructure provisioned: 2 nodes for production'
null_resource.provision_note: Creation complete after 0s [id=5542324021968538359]

```

Apply complete! Resources: 3 added, 0 changed, 0 destroyed.

Outputs:

```

ansible_inventory_path = "../ansible/inventory.ini"
environment = "production"
vm_count = 2

```

PS C:\Users\Admin\Downloads\sre-project\sre-project\terraform>

Main.tf

```

main.tf M X
terraform > main.tf > terraform
1 terraform {
2   required_version = ">= 1.0"
3   required_providers {
4     # Uncomment for AWS:
5     # aws = {
6     #   source = "hashicorp/aws"
7     #   version = "~> 5.0"
8     # }
9     local = {
10      source = "hashicorp/local"
11      version = "~> 2.4"
12    }
13    null = {
14      source = "hashicorp/null"
15      version = "~> 3.2"
16    }
17  }
18 }
19
20 #Variables
21 variable "environment" {
22   description = "Environment name"
23   type        = string
24   default     = "production"
25 }
26
27 variable "app_name" {
28   description = "Application name"
29   type        = string
30   default     = "sre-app"
31 }
32
33 variable "vm_count" {
34   description = "Number of worker nodes"
35   type        = number
36   default     = 2
37 }
38
39 # Local: simulate VM provisioning
40 # (Replace this section with AWS/GCP/Azure provider for cloud)
41

```

```

41
42 resource "local_file" "inventory" {
43   content = templatefile("${path.module}/templates/inventory.tpl", {
44     app_name = var.app_name
45     environment = var.environment
46     vm_count = var.vm_count
47   })
48   filename = "${path.module}/../ansible/inventory.ini"
49 }
50
51 resource "local_file" "env_config" {
52   content = jsonencode({
53     environment = var.environment
54     app_name = var.app_name
55     vm_count = var.vm_count
56     db_host = "postgres"
57     redis_host = "redis"
58     created_at = timestamp()
59   })
60   filename = "${path.module}/../ansible/vars/terraform_output.json"
61 }
62
63 resource "null_resource" "provision_note" {
64   triggers = {
65     always_run = timestamp()
66   }
67
68   provisioner "local-exec" {
69     command = "echo 'Infrastructure provisioned: ${var.vm_count} nodes for ${var.environment}'"
70   }
71 }
72
73 # Outputs
74 output "environment" {
75   value = var.environment
76 }
77
78 output "vm_count" {
79   value = var.vm_count
80 }

```

```

68     provisioner "local-exec" {
69         command = "echo 'Infrastructure provisioned: ${var.vm_count} nodes for ${var.environment}'"
70     }
71 }
72
73 # Outputs
74 output "environment" {
75     value = var.environment
76 }
77
78 output "vm_count" {
79     value = var.vm_count
80 }
81
82 output "ansible_inventory_path" {
83     value = local_file.inventory.filename
84 }
85

```

7. Docker Swarm

```

PS C:\Users\Admin\Downloads\sre-project\sre-project> docker swarm init
Swarm initialized: current node (zjzhbax9bcqu5yu9izb0sy74t) is now a manager.

To add a worker to this swarm, run the following command:

    docker swarm join --token SWMTKN-1-0lc78wvocsjm4qqbqqigz2it34sxxh115jaeljy5wtet96ojkcd-b6wh8phmknig3kstr7181ioo2 192.168.65.3:2377

To add a manager to this swarm, run 'docker swarm join-token manager' and follow the instructions.

```

```

PS C:\Users\Admin\Downloads\sre-project\sre-project> docker stack deploy -c swarm-stack.yml sre
Since --detach=false was not specified, tasks will be created in the background.
In a future release, --detach=false will become the default.
Creating network sre_app-network
Creating service sre_auth
Creating service sre_product
Creating service sre_order
Creating service sre_payment
Creating service sre_prometheus
Creating service sre_postgres
Creating service sre_grafana
Creating service sre_user-profile
Creating service sre_frontend
Creating service sre_notification
PS C:\Users\Admin\Downloads\sre-project\sre-project>

```

```

PS C:\Users\Admin\Downloads\sre-project\sre-project> docker service ls

```

ID	NAME	MODE	REPLICAS	IMAGE	PORTS
16ralg1iqg2a	sre_auth	replicated	0/2	sre-auth:latest	*:5001->5001/tcp
il028jjnb7w0	sre_frontend	replicated	0/1	sre-frontend:latest	*:80->80/tcp
ao9jv7dbdqpg	sre_grafana	replicated	0/1	grafana/grafana:latest	*:3000->3000/tcp
02z4xqvn228d	sre_notification	replicated	0/1	sre-notification:latest	*:5005->5005/tcp
3qiv7r5a6t1j	sre_order	replicated	0/3	sre-order:latest	*:5003->5003/tcp
s8s2z2ijqjag	sre_payment	replicated	0/3	sre-payment:latest	*:5004->5004/tcp
bcfk6jy2djpg	sre_postgres	replicated	0/2	postgres:15-alpine	
0ryf8zmall07	sre_product	replicated	1/1	sre-product:latest	*:5002->5002/tcp
u2h9bhtyxq8m	sre_prometheus	replicated	1/1	prom/prometheus:latest	*:9090->9090/tcp
sf9zgwwhcm58	sre_user-profile	replicated	0/2	sre-user-profile:latest	*:5006->5006/tcp

8. Kubernetes

```

service/frontend created
PS C:\Users\Admin\Downloads\sre-project\sre-project> kubectl get pods -n sre-app
NAME                                READY   STATUS              RESTARTS   AGE
auth-6f86cb696d-m615d              0/1     ErrImageNeverPull    0           20s
auth-6f86cb696d-t45ct              0/1     ErrImageNeverPull    0           20s
frontend-57fd48d7d9-v29lw          0/1     ErrImageNeverPull    0           19s
notification-575fb697d4-4hbns      0/1     ErrImageNeverPull    0           20s
order-7dd44bbbdd-8t5rd              0/1     ErrImageNeverPull    0           20s
order-7dd44bbbdd-z8bzq              0/1     ErrImageNeverPull    0           20s
payment-55584b9b58-jw2kz           0/1     ErrImageNeverPull    0           20s
payment-55584b9b58-sffpq           0/1     ErrImageNeverPull    0           20s
product-588f47885d-knshv            0/1     ErrImageNeverPull    0           20s
product-588f47885d-q4r9v           0/1     ErrImageNeverPull    0           20s
user-profile-54dc645f56-h6gqr       0/1     ErrImageNeverPull    0           20s
user-profile-54dc645f56-wwtmm      0/1     ErrImageNeverPull    0           20s
PS C:\Users\Admin\Downloads\sre-project\sre-project>

```

```

PS C:\Users\Admin\Downloads\sre-project\sre-project> kubectl apply -f kubernetes/deployments.yml
deployment.apps/auth created
service/auth created
deployment.apps/product created
service/product created
deployment.apps/order created
service/order created
deployment.apps/payment created
service/payment created
deployment.apps/notification created
service/notification created
deployment.apps/user-profile created
service/user-profile created
configmap/app-config created
horizontalpodautoscaler.autoscaling/order-hpa created
deployment.apps/frontend created
service/frontend created

```

9.Capacity Planning

curl http://localhost:5001/health


```

StatusCode      : 200
StatusDescription : OK
Content         : {"service":"auth","status":"healthy"}

RawContent      : HTTP/1.1 200 OK
                  Connection: close
                  Content-Length: 38
                  Content-Type: application/json
                  Date: Sun, 17 May 2026 19:16:07 GMT
                  Server: Werkzeug/3.1.8 Python/3.11.15

                  {"service":"auth","status":"health...
Forms           : {}
Headers         : {[Connection, close], [Content-Length, 38], [Content-Type, application/json], [Date, Sun, 17 May
                  2026 19:16:07 GMT]...}
Images          : {}
InputFields     : {}
Links           : {}
ParsedHtml      : mshtml.HTMLDocumentClass
RawContentLength : 38

```

curl <http://localhost:5002/products>

```

StatusCode      : 200
StatusDescription : OK
Content         : {"products":[{"id":1,"name":"Laptop","price":999.99,"stock":50},{"id":2,"name":"Phone","price":599.
                  99,"stock":100},{"id":3,"name":"Tablet","price":399.99,"stock":75}], "total":3}

RawContent      : HTTP/1.1 200 OK
                  Connection: close
                  Content-Length: 178
                  Content-Type: application/json
                  Date: Sun, 17 May 2026 19:16:47 GMT
                  Server: Werkzeug/3.1.8 Python/3.11.15

                  {"products":[{"id":1,"name":"Lapt...
Forms           : {}
Headers         : {[Connection, close], [Content-Length, 178], [Content-Type, application/json], [Date, Sun, 17 May
                  2026 19:16:47 GMT]...}
Images          : {}
InputFields     : {}
Links           : {}
ParsedHtml      : mshtml.HTMLDocumentClass
RawContentLength : 178

```

curl <http://localhost:5003/orders>

```

StatusCode      : 200
StatusDescription : OK
Content         : {"orders":[],"total":0}

RawContent      : HTTP/1.1 200 OK
                  Connection: close
                  Content-Length: 24
                  Content-Type: application/json
                  Date: Sun, 17 May 2026 19:17:31 GMT
                  Server: Werkzeug/3.1.8 Python/3.11.15

                  {"orders":[],"total":0}

Forms           : {}
Headers         : {[Connection, close], [Content-Length, 24], [Content-Type, application/json], [Date, Sun, 17 May
                  2026 19:17:31 GMT]...}
Images          : {}
InputFields     : {}
Links           : {}
ParsedHtml      : mshtml.HTMLDocumentClass
RawContentLength : 24

```

curl <http://localhost:5004/health>

```

StatusCode      : 200
StatusDescription : OK
Content         : {"service":"payment","status":"healthy"}

RawContent      : HTTP/1.1 200 OK
                  Connection: close
                  Content-Length: 41
                  Content-Type: application/json
                  Date: Sun, 17 May 2026 19:18:02 GMT
                  Server: Werkzeug/3.1.8 Python/3.11.15

                  {"service":"payment","status":"hea...
Forms           : {}
Headers         : {[Connection, close], [Content-Length, 41], [Content-Type, application/json], [Date, Sun, 17 May
                  2026 19:18:02 GMT]...}
Images          : {}
InputFields     : {}
Links           : {}
ParsedHtml      : mshtml.HTMLDocumentClass
RawContentLength : 41

```

curl <http://localhost:5005/health>

```

StatusCode      : 200
StatusDescription : OK
Content         : {"service":"notification","status":"healthy"}

RawContent      : HTTP/1.1 200 OK
                  Connection: close
                  Content-Length: 46
                  Content-Type: application/json
                  Date: Sun, 17 May 2026 19:18:18 GMT
                  Server: Werkzeug/3.1.8 Python/3.11.15

                  {"service":"notification","status"...
Forms           : {}
Headers         : {[Connection, close], [Content-Length, 46], [Content-Type, application/json], [Date, Sun, 17 May
                  2026 19:18:18 GMT]...}
Images          : {}
InputFields     : {}
Links           : {}
ParsedHtml      : mshtml.HTMLDocumentClass
RawContentLength : 46

```

curl <http://localhost:5006/health>

```

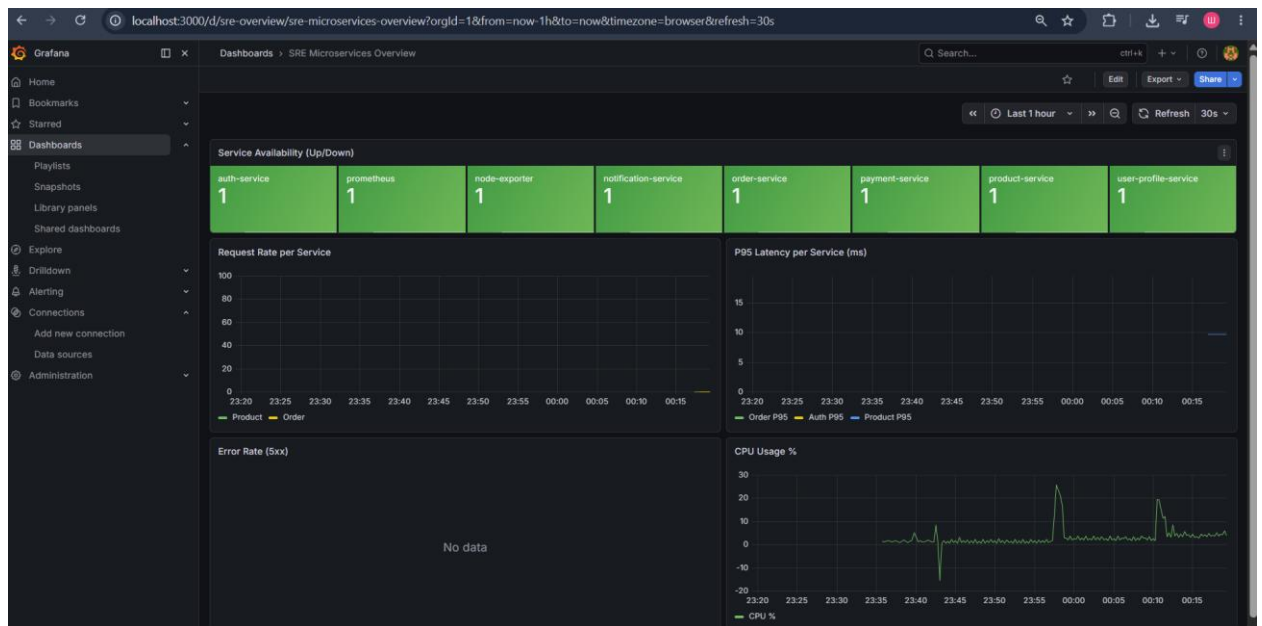
StatusCode      : 200
StatusDescription : OK
Content         : {"service":"user-profile","status":"healthy"}

RawContent      : HTTP/1.1 200 OK
                  Connection: close
                  Content-Length: 46
                  Content-Type: application/json
                  Date: Sun, 17 May 2026 19:18:43 GMT
                  Server: Werkzeug/3.1.8 Python/3.11.15

                  {"service":"user-profile","status"...
Forms           : {}
Headers         : {[Connection, close], [Content-Length, 46], [Content-Type, application/json], [Date, Sun, 17 May
                  2026 19:18:43 GMT]...}
Images          : {}
InputFields     : {}
Links           : {}
ParsedHtml      : mshtml.HTMLDocumentClass
RawContentLength : 46

```

Grafana with load data



CI/CD Pipeline

